

LAB 5

1. Write a Java program that reads a string from the user and uses StringTokenizer to split the string into individual words. Print each word on a new line.

PROGRAM:

```
package Lab5;

import java.util.Scanner;
// Import StringTokenizer from java.util
import java.util.StringTokenizer;

public class TokenizerWords {

    public static void main(String[] args) {
        // Scanner object to take string input
        Scanner sc1 = new Scanner(System.in);
        System.out.println("Enter a string: ");
        String str1 = sc1.nextLine();

        // StringTokenizer object to define string and delimiter
        StringTokenizer st = new StringTokenizer(str1, " ");
        // Loop to iterate through string if it has more than one tokens
        while (st.hasMoreTokens()) {
            // printing tokens or words in the array
            System.out.println(st.nextToken());
        }
        sc1.close();
    }
}
```

OUTPUT:



```
Enter a string:
My name is ashi
My
name
is
ashi
```

2. Write a Java program that reads a string from the user and uses StringTokenizer to count the number of words in the string.

PROGRAM:

```
package Lab5;

import java.util.Scanner;
// Import StringTokenizer from java.util
import java.util.StringTokenizer;

public class TokenizerNumberOfWords {

    public static void main(String[] args) {
        // Scanner object to take string input
```

```

Scanner sc1 = new Scanner(System.in);
System.out.println("Enter a string: ");
String str1 = sc1.nextLine();

int count = 0;
// StringTokenizer object to define string and delimiter
StringTokenizer st = new StringTokenizer(str1, " ");

// Loop to iterate through string if it has more than one tokens
while (st.hasMoreTokens()) {
    // Split string into tokens
    st.nextToken();
    count++;
}

System.out.println("The number of words is : " + count);
sc1.close();
}
}

```

OUTPUT:

```

<terminated> TokenizerNumberOfWords [Java Application] C:\Use
Enter a string:
My name is ashi
The number of words is : 4

```

3. Write a Java program to create a LinkedList of strings, add elements at specific positions (beginning, middle, end), and print the list.

PROGRAM:

```

package Lab5;
import java.util.LinkedList;      // Importing LinkedList Class

public class StringLinkedList {

    public static void main(String[] args) {
        // Creating LinkedList object of String type to store animal names
        LinkedList<String> animalLL = new LinkedList<String>();

        // Adding animal names in the LinkedList
        animalLL.add("dog");
        animalLL.add("cat");
        animalLL.add("horse");
        animalLL.add("sheep");
        animalLL.add("cow");
        animalLL.add("mouse");

        // Printing the initial LinkedList
        System.out.println("Animal name string LinkedList : \n" + animalLL);

        // Adding at first
        animalLL.addFirst("monkey");

        // Printing LinkedList After adding an element at first
        System.out.println("Animal LinkedList after adding at first
position: \n" + animalLL);
    }
}

```

```

        // To add element in the middle
        // Storing size of LinkedList in a variable
        int s = animalLL.size();

        // If the size of LinkedList is even then add element after the half
of the LinkedList size
        // Else add element at the middle element
        if (s%2==0) {
            animalLL.add(((s/2)+1), "parrot");
        } else {
            animalLL.add((s+1)/2, "peacock");
        }
        System.out.println("Animal LinkedList after adding at middle : \n "
+ animalLL);

        // To add element at last
        animalLL.addLast("whale");

        // Printing after adding element at last
        System.out.println("Animal LinkedList after adding at last position:
\n" + animalLL);
    }
}

```

OUTPUT:

```

<terminated> StringLinkedList [Java Application] C:\Users\Ashi Choudhary\p2\p001\plugi
Animal name string LinkedList :
[dog, cat, horse, sheep, cow, mouse]
Animal LinkedList after adding at first position:
[monkey, dog, cat, horse, sheep, cow, mouse]
Animal LinkedList after adding at middle :
[monkey, dog, cat, horse, peacock, sheep, cow, mouse]
Animal LinkedList after adding at last position:
[monkey, dog, cat, horse, peacock, sheep, cow, mouse, whale]

```

4. Write a Java program to sort a given array list.

PROGRAM:

```

package Lab5;
import java.util.Arrays;
import java.util.Collections;

public class ArraySortDemo {

    public static void main(String[] args) {
        Integer age[] = {10, 2, 7, 45, 30};
        String name[] = {"ashi", "aryan", "suhani", "dinesh", "rajesh"};

        // Descending order sorting
        Arrays.sort(age, Collections.reverseOrder());
        Arrays.sort(name, Collections.reverseOrder());

        System.out.println("Age array sorted in descending order : ");
    }
}

```

```

        System.out.println(Arrays.toString(age));

        System.out.println("Name array sorted in descending order : ");
        System.out.println(Arrays.toString(name));

        System.out.println();
        // Integers - Ascending order

        int num1[] = {8, 4, 3,5,6};
        int num2[] = {10, 7, 9, 13, 17 };

        Arrays.sort(num1);
        Arrays.sort(num2);
        System.out.println("num1 array sorted in descending order : ");
        System.out.println(Arrays.toString(num1));

        System.out.println("num2 array sorted in descending order : ");
        System.out.println(Arrays.toString(num2));

        // String - ascending order
        String str1[] = {"C", "O", "I", "P", "U"};
        String str2[] = {"E", "H", "G", "A", "C"};

        Arrays.sort(str1);
        Arrays.sort(str2);

        System.out.println("str1 array sorted in ascending order : ");
        System.out.println(Arrays.toString(str1));

        System.out.println("str2 array sorted in ascending order : ");
        System.out.println(Arrays.toString(str2));
    }
}

```

OUTPUT:

```

Age array sorted in descending order :
[45, 30, 10, 7, 2]
Name array sorted in descending order :
[suhani, rajesh, dinesh, ashi, aryan]

num1 array sorted in descending order :
[3, 4, 5, 6, 8]
num2 array sorted in descending order :
[7, 9, 10, 13, 17]
str1 array sorted in ascending order :
[C, I, O, P, U]
str2 array sorted in ascending order :
[A, C, E, G, H]

```

5. Write a Java program to replace the second element of an ArrayList with the specified element.

PROGRAM:

```

package Lab5;
import java.util.*;      // Importing util and all its packages
public class ArrayListReplaceDemo {

    public static void main(String[] args) {
        // Creating an arrayList object of List Interface class ArrayList
        List<Integer> arrList = new ArrayList<Integer>();

        // Adding elements in the arrayList
        arrList.add(10);
        arrList.add(20);
        arrList.add(40);
        arrList.add(55);
        arrList.add(20);
        arrList.add(100);
        arrList.add(34);

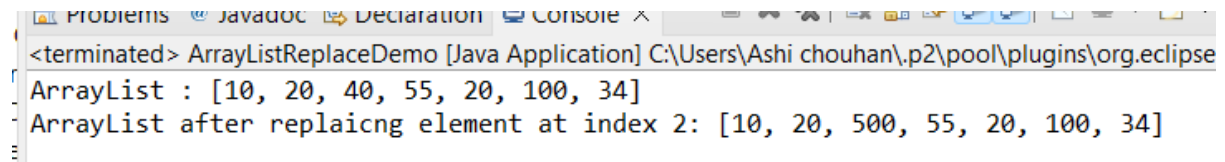
        // Printing the arrayList
        System.out.println("ArrayList : " + arrList);

        // Replacing the element at index 2 with value 500
        arrList.set(2, 500);

        // Printing array after replacement
        System.out.println("ArrayList after replaicng element at index 2: "
+ arrList);
    }
}

```

OUTPUT:



```

<terminated> ArrayListReplaceDemo [Java Application] C:\Users\Ashi chouhan\.p2\pool\plugins\org.eclipse
ArrayList : [10, 20, 40, 55, 20, 100, 34]
ArrayList after replaicng element at index 2: [10, 20, 500, 55, 20, 100, 34]

```

6. Write a Java program to iterate a linked list in reverse order.

PROGRAM:

```

package Lab5;

import java.util.Collections;      // Importing Collections Superclass
import java.util.LinkedList;      // Importing LinkedList class

public class LinkedListReverseOrder {

    public static void main(String[] args) {
        // Creating a LinkedList of string type for storing names of
students
        LinkedList<String> studLinkedList = new LinkedList<String>();
        // Adding student names in the LinkedList studLinkedList
        studLinkedList.add("ashi");
        studLinkedList.add("aryan");
        studLinkedList.add("suhani");
        studLinkedList.add("ranjesh");
    }
}

```

```

studLinkedList.add("dinesh");
studLinkedList.add("nikhil");
studLinkedList.add("adit");
studLinkedList.add("saroj");

// Printing the linkedList
System.out.println("Students LinkedList : " + studLinkedList);

// Sorting the linkedList in Reverse order
Collections.sort(studLinkedList, Collections.reverseOrder());

// Printing LinkedList after sorting in reverse order
System.out.println("Students LinkedList in reverse order : " +
studLinkedList);
    }
}

```

OUTPUT:

```

<terminated> LinkedListReverseOrder [Java Application] C:\Users\Ashi chouhan\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.fu
Students LinkedList : [ashi, aryan, suhani, ranjesh, dinesh, nikhil, adit, saroj]
Students LinkedList in reverse order : [suhani, saroj, ranjesh, nikhil, dinesh, ash, aryan, adit]

```

7. Write a Java program to retrieve, but not remove, the last element of a linked list.

PROGRAM:

```

package Lab5;
import java.util.LinkedList; // Importing LinkedList class
public class retrieveLastLinkedList {

    public static void main(String[] args) {
        // Creating LinkedList Object of integer type
        LinkedList<Integer> intLL = new LinkedList<Integer>();

        // Adding elements in the LinkedList
        intLL.add(2);
        intLL.add(5);
        intLL.add(1);
        intLL.add(7);
        intLL.add(8);
        intLL.add(9);

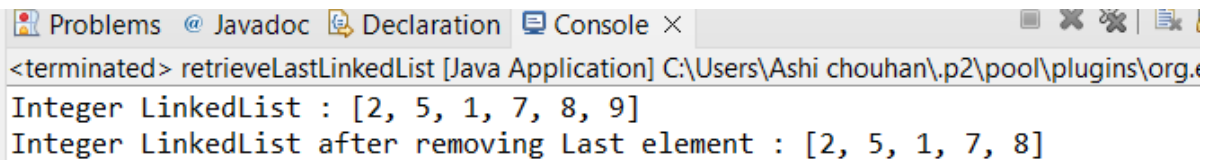
        // Printing the initial linkedList
        System.out.println("Integer LinkedList : " + intLL);

        // Removing the last element From the LinkedList
        intLL.removeLast();

        // Printing the LinkedList after removing last Element
        System.out.println("Integer LinkedList after removing Last element :
" + intLL);
    }
}

```

OUTPUT:



```
<terminated> retrieveLastLinkedList [Java Application] C:\Users\Ashi chouhan\p2\pool\plugins\org.e
Integer LinkedList : [2, 5, 1, 7, 8, 9]
Integer LinkedList after removing Last element : [2, 5, 1, 7, 8]
```

8. Write a Java program to create a LinkedList of integers and print all the elements.

PROGRAM:

```
package Lab5;
import java.util.LinkedList;    // Importing LinkedList class

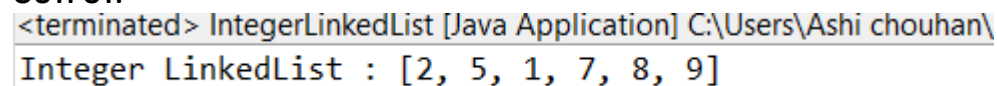
public class IntegerLinkedList {

    public static void main(String[] args) {
        LinkedList<Integer> intLL = new LinkedList<Integer>();

        // Adding elements in the LinkedList
        intLL.add(2);
        intLL.add(5);
        intLL.add(1);
        intLL.add(7);
        intLL.add(8);
        intLL.add(9);

        // Printing the initial linkedList
        System.out.println("Integer LinkedList : " + intLL);
    }
}
```

OUTPUT:



```
<terminated> IntegerLinkedList [Java Application] C:\Users\Ashi chouhan\
Integer LinkedList : [2, 5, 1, 7, 8, 9]
```